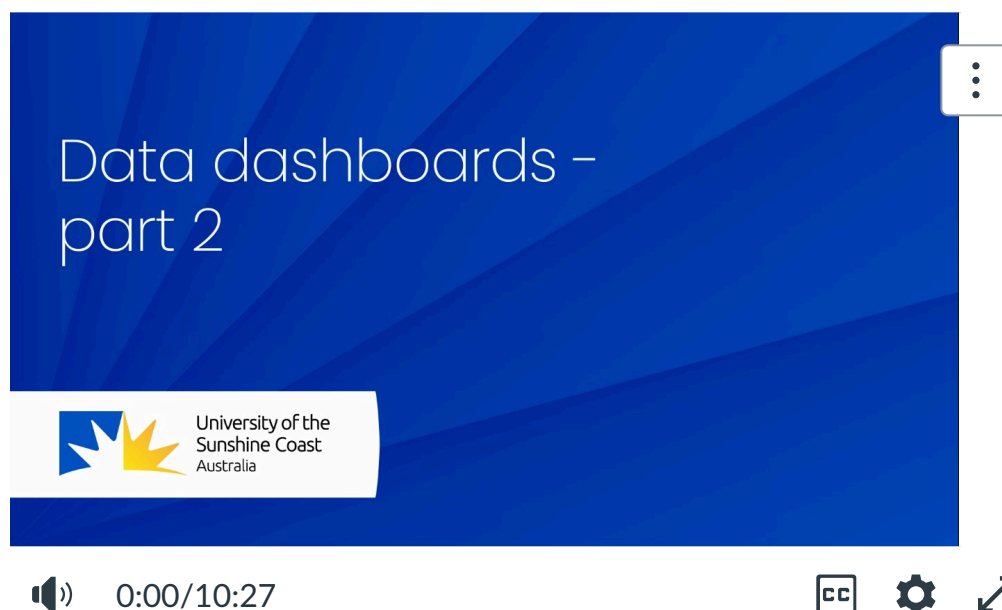


9.3 Data dashboard design ↕

□ Watch: Data dashboards (part 2) [10.27 mins]

Watch the video below, which explains the principles of data dashboard design and development that are used in the process of creating useful and appropriate data dashboards.

This is useful as understanding data dashboard design, which includes knowing the purpose of a data dashboard and considering the needs of the dashboard's users, results in better dashboards and ultimately improved decision making.



[Transcript](#)

<https://learn.usc.edu.au/courses/35242/files/3138510?wrap=1>

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Data dashboards \(part 2\)](https://learn.usc.edu.au/courses/35242/files/3138673?wrap=1)
<https://learn.usc.edu.au/courses/35242/files/3138673?wrap=1>

The most important consideration in dashboard design is that the data dashboard supports the objectives of the business or business function for which it is being developed. It must meet the needs of the end user/s. Nevertheless, there are several elements that are common in successful dashboard design.

Understanding the purpose of the data dashboard

You must consider what needs to be accomplished through the use of the dashboard.

Considering the needs of the end users

You must design the dashboard to support a particular user or group of users in the task that they need to do. Therefore, the dashboard designer must understand both the needs of the end users and how addressing those needs will support the organisation's dashboard objectives.

Data dashboard engineering

The end users must be involved in the process of developing the data dashboard, as all the information that is displayed on a dashboard must be relevant to supporting their needs.

Design principles

Design principles must be taken into consideration. The dashboard must consider:

- if it is easy to read and interpret
- if it is too complicated/overly complex
- if it is too sparse
- the environment it is to be used in
- the device that will be used to access it
- the mode of user interaction (e.g. touchscreen, projector)
- the context of the information to be displayed
- if users will need to do further analysis and how this may be facilitated
- the data-ink ratio and minimising eye travel
- the maintenance of the dashboard, including the skills of those responsible for this
- the dashboard's effectiveness
- any future modifications to the dashboard if objectives change
- the ability to test with end users for user experience
- the ability to test the dashboard prior to implementation.

There must also be regular reviews of the dashboard with management and/or end users.

□ Watch: Dashboard design best practices [4.10 mins]

This video is useful as it provides a checklist to consider before designing a dashboard. It also explains some do's and don'ts that can help in the creation of practical business dashboards.

As you watch, consider:

- How could these tips help you design the dashboard for [Task 2](https://learn.usc.edu.au/courses/35242/pages/task-2-data-visualisation-report-and-creative-artefact) (<https://learn.usc.edu.au/courses/35242/pages/task-2-data-visualisation-report-and-creative-artefact>)?
- Why is building clear and easy-to-understand business dashboards important?



Source: © Geckoboard (2020).

Common mistakes in data dashboard design

There are a number of commonly made mistakes in data dashboard design. These include:

- a lack of consideration for the organisation's reasons for developing a dashboard
- not obtaining sufficient input from the end users
- not taking account of the environment in which the dashboard will be used
- inappropriate placement of complementary elements on the dashboard
- using inappropriate types of visualisation (charts, tables, etc.)
- not using principles of good design on individual dashboard components
- using an unattractive or cluttered visual display
- not considering the end users' potential future needs.

□ Activity: Check your understanding of common mistakes in dashboard design

Can you identify the mistakes in the images below? Click on the arrows to move through each image and make a list of errors. Click on the solutions tab below to see if you were on the right track.



Source: Matillion (2020).

□ Watch: Five dashboard design tips [6.21 mins]

This video is useful as it explains that a poorly designed dashboard will increase the cognitive load for the audience and reflect poorly on the designer. It may also mislead and lead to poor decision making. The video describes five common mistakes that should be avoided.

As you watch, consider:

- How does each “mistake” impact the audience?

- Why can the design of a dashboard make or break how well your message is communicated?

Source: © MyOnlineTrainingHub (2022).

□ Activity: The crime rate perception gap

Reflect on this webpage about the crime rate perception gap in the United States. Note the webpage was published in February 2019.

As you look at the visualisation, consider and make notes on:

- the type of audience that this data dashboard would be useful for
- whether the message would be clear without the text explanation
- what improvements could be made.

Explain your reasoning and bring your notes to the workshop.

Routley, N. (2019, February 5). [The crime rate perception gap](https://www.visualcapitalist.com/crime-rate-perception-gap/)  (<https://www.visualcapitalist.com/crime-rate-perception-gap/>). *Visual Capitalist*.